INCH-POUND

MIL-S-20708/74E
13 March 1989

SUPERSEDING
MIL-S-20708/74D
30 July 1987

MILITARY SPECIFICATION SHEET
SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 37TRX6A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I Requirements.

Requirement	Value	Unit	Tolerance
Frequency	60	Hz	±1%
Primary Voltage	115	volts	±1%
Primary Current	830.0	milliamps	maximum
Primary Power	9.25	watts	maximum
Impedance			
Zss	7.0-10.0	ohms	min-max
Impedance Angle			
Zss	30.0-40.0	degrees	min-max
Transformation Ratio	0.783	-----	±2%
Phase Shift (Lead)	3.5	degrees	maximum
Electrical Error	10.0	minutes	maximum
Receiver Error	36.0	minutes	maximum
Null Voltage			
Total	170.0	millivolts	maximum
Fundamental	35.0	millivolts	maximum
Synchronizing Time 30° ±2°	1.0	seconds	maximum
Synchronizing Time 177° ±2°	2.0	seconds	maximum
Torque Gradient	0.90	ounce-inches per degree	minimum
Radial Play	0.0008	inches	maximum
End Play	0.0030	inches	±0.0020
Temperature Rise	35.0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq (-5%)	18.0	watts	maximum

This unit shall not be subjected to the altitude requirements of MIL-S-20708.

E denotes changes

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TABLE II Military part number variant characteristics

Military Part No 1/	A ± .010	B ± .010	G Maximum	Shaft Style	Terminal End	U +.0000 -.0002
M20708/74-01A	.700	.730	4.030	Splined	Terminal Board	2405

(5)

1/Military part number M20708/74-01A shall be the NATO standard

1/The suffix letter following the numerical base number corresponds to the latest modification letter in the type designation

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/74C	MIL-S-20708/74B	
M20708/74C-001	M20708/74-001	M20708/74-01A

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification. Part number changes do not affect form, fit, or function of the synchros listed therein.

Custodians
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI, AV
Navy-OS
Air Force-11, 17, 99
DLA-ES

User Activities
Navy-MC, SG

Preparing Activity
Navy AS

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